

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	Hard

Question

A scientist studying the life cycle of dragonflies counted the number of dragonflies in a certain habitat each day for **46** days. On February **15**, there were **99** dragonflies in the habitat. The percent increase in the number of dragonflies in the habitat from January **1** to February **15** was **12.50%**. How many dragonflies were in the habitat on January **1**?

Answer

- A. **88**
- B. **87**
- C. **12**
- D. **8**

Correct Answer: A

Rationale

Choice A is correct. It's given that a scientist studying the life cycle of dragonflies counted the number of dragonflies in a certain habitat each day for **46** days. It's also given that on February **15**, there were **99** dragonflies in the habitat and that the percent increase in the number of dragonflies in the habitat from January **1** to February **15** was **12.50%**. This can be represented by the equation $99 = \left(1 + \frac{12.50}{100}\right)x$, where x represents the number of dragonflies in the habitat on January **1**. This equation can be rewritten as $99 = 1.125x$. Dividing both sides of this equation by **1.125** yields $88 = x$. Therefore, there were **88** dragonflies in the habitat on January **1**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.